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Materials and Methods - Entrainment fluxes

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The entrainment fluxes in a shear-free convective boundary layer were studied in this experiment. Elevated Richardson number entrainment fluxes properties were of interest, specifically from 40 to 190 Ri. These kind of fluxes are those where one fluid is transported and mixed with another one.

A saline container was used in this experiment to manipulate the border conditions and to achieve the conservation of the homogeneity of the surface and external forces. The fixation of some parameters to study others was achieved by the usage of the tank.

1 Materials

The experimental set up included a 1 m x 1 m glass container with a height of 0.5 m. A scanning rail which attached the 2.5 W illuminating laser connected with an optical fiber. A digital 10-bit camera that recorded the fluorescence images in front of the vessel. A tray of 49 porous tubes (Gardenia, Ulm, Germany, external diameter 17 mm) were used to supply fresh water, which was kept in an external tank, above the first one. The surface of the porous tubes was evened with a meticulously leveled bed of aquarium quartz pebbles. Fluorescent dye of 30 known concentrations of disodium fluorescein ($C_{20}H_{10}O_5Na_2$) was also used for the experiment.

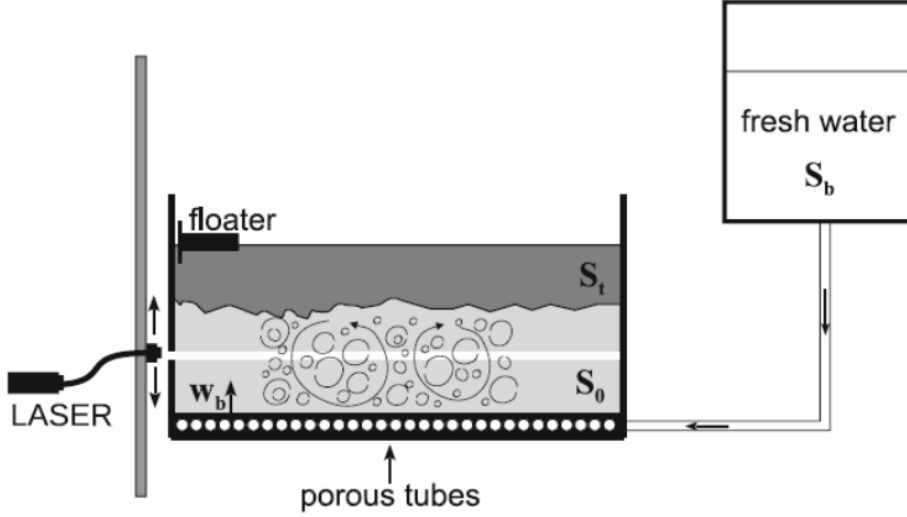


Figure 1: Skeeth of the saline convection container set-up.

2 Material preparation and set up

The first step was to fill the tank through the porous tubes with well-mixed water in order to start with a salinity of S_0 until the height was of $h_0 \approx 0.1$ m. Subsequently, a lower salinity layer of S_t was added on top of the mixing layer so a buoyancy jump of $\Delta_b = g[S_0 - S_t]$ was created. The higher reservoir contained one of the fluorescents dyes, so planar laser induced fluorescence (Ferrier et al., 1993; Snyder et al., 2002; Dop et al., 2005) allowed to measure the dye concentration in the mixed layer, which was illuminated by the laser sheet. A slow filling procedure was used by dripping the layer of water onto four floating devices submerged just below the water surface to ensure lateral flow. About five minutes were necessary to make each layer-concentration measurement.

Two types of images were recorded by the camera: one of the mixed layer only (image type I), and another showing both the top layer and the mixed layer (image type II). The camera was triggered by the position of the fiber on the transport rail (Figure 1). The measurement of the mixed concentrations without interference of the high concentrations in the upper layer was allowed by type I images. This was achieved by setting the camera shutter to close before the beam illuminated the top layer. A significant amount of information on the structure of the inversion region and the interacting convection dynamics was provided by type II images, as well as the inflow velocity ω_b by tracking the evolution in time of the height of the total water column.

To take the images a piece of foam attached to a vertical plate was placed so a floating device on top of the water was produced (Figure 1). The beam was blocked by a small vertical plate after passing the floater, marking the water level in the camera image. The inflow of ω_b provoked a rise in water level, so the floater rised at the same speed. Very accurate estimations of ω_b were achieved by the location of the laser-blocking floater.

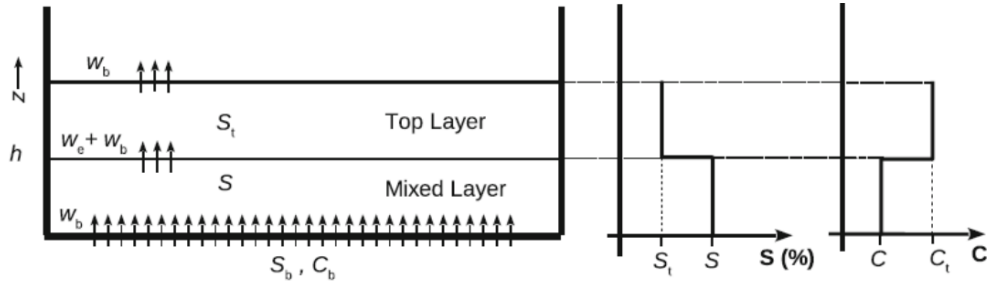


Figure 2: Schematic view of the two layers separated by a buoyancy jump (left) and corresponding salinity and dye concentration profiles (right). The top water level increases by the inflow rate ω_b , whereas the mixed layer also grows by entrainment, i.e. at rate: $\omega_e + \omega_b$.

3 References

Ferrier AJ, Funk DR, Roberts PJW (1993) Application of optical techniques to the study of plumes in stratified fluids. *Dyn Atmos Oceans* 20:155-183.

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